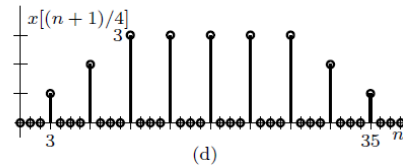
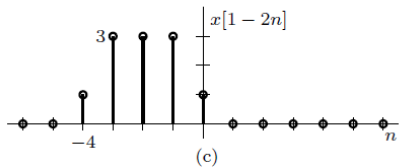
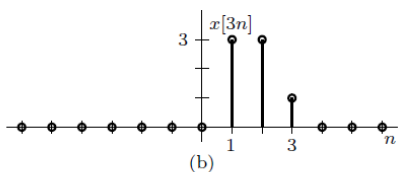
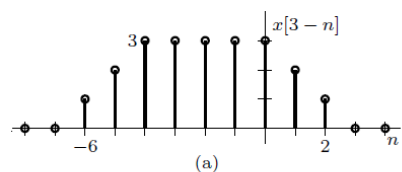


Answer Keys to HW 3

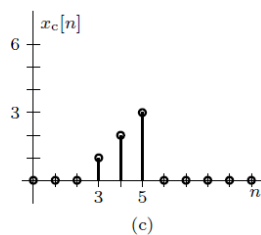
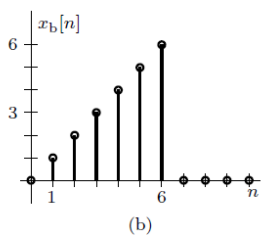
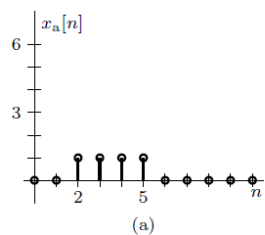
1)



2)

$$\theta = \Omega N \quad \text{for any integer } N.$$

3)



4)

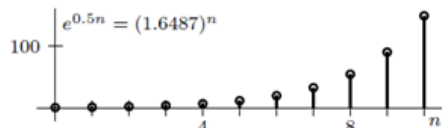
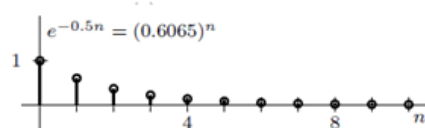
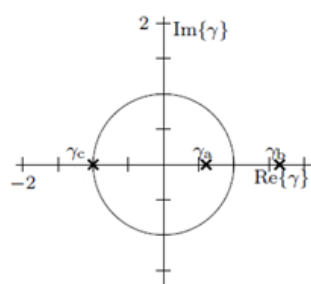
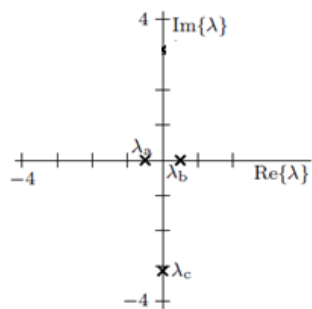
$$x_a[n] = (n+3)(u[n+3] - u[n]) + (3-n)(u[n] - u[n-4]).$$

$$x_a[n] = (n+3)(u[n+2] - u[n]) + (3-n)(u[n] - u[n-3]).$$

$$x_c[n] = 2n(u[n] - u[n-4]) + 2n(u[-n] - u[-n-4]).$$

$$x_c[n] = 2n(u[n+3] - u[n-4]).$$

5)



- (a) Since $e^{-0.5n} = (0.6065)^n$, $\lambda_a = -0.5$ and $\gamma_a = 0.6065$,
 (b) Since $e^{0.5n} = (1.6487)^n$, $\lambda_b = 0.5$ and $\gamma_b = 1.6487$,
 (c) Since $e^{-j\pi n} = (-1)^n$, $\lambda_c = -j\pi$ and $\gamma_c = -1$,
 (d) Since $e^{-(1+j\pi)n} = (-\frac{1}{e})^n = (-0.3679)^n$, $\lambda_a = -(1+j\pi)$ and $\gamma_a = -\frac{1}{e} = -0.3679$,
 (e) Since $e^{(1-j\frac{\pi}{3})n} = (1.3591 - 2.3541j)^n$, $\lambda_f = 1 - j\frac{\pi}{3}$ and $\gamma_f = 1.3591 - 2.3541j$,

6)

$$\omega_0 = 60\pi + 200\pi k.$$

the three smallest values of $|\omega_0|$ are 60π , 140π , and 260π rad/s.

7)

From the perspective of apparent frequency, exponentials are no different than sinusoids. Thus, we compute the apparent frequency of each complex exponential as $\Omega_a = \langle \Omega + \pi \rangle_{2\pi} - \pi$. This provides frequencies in the fundamental range, $-\pi \leq \Omega_a < \pi$. Frequencies in the range $0 \leq \Omega < 2\pi$ can be obtained by adding 2π to all values $\Omega_a < 0$.

- (a) $e^{j(0.2\pi n + \theta)}$.
 (b) $e^{0n} = 1$.
 (c) $e^{-j1.95n}$. in the range $0 \leq \Omega < 2\pi$, $e^{j4.3332n}$.
 (d) $e^{-j0.7\pi n}$. in the range $0 \leq \Omega < 2\pi$, $e^{j1.3\pi n}$.

8)

a)

$$x(nT) = 10 \cos(2\pi 1000nT) + 3.1623 \cos(2\pi 1500nT - 1.1071).$$

$$x[n] = 10 \cos\left(\frac{\pi}{2}n\right) + 3.1623 \cos\left(\frac{3\pi}{4}n - 1.1071\right).$$

b) $T < 1/5000$